



The Electric Field and Potential of a Homogeneously Charged Sphere

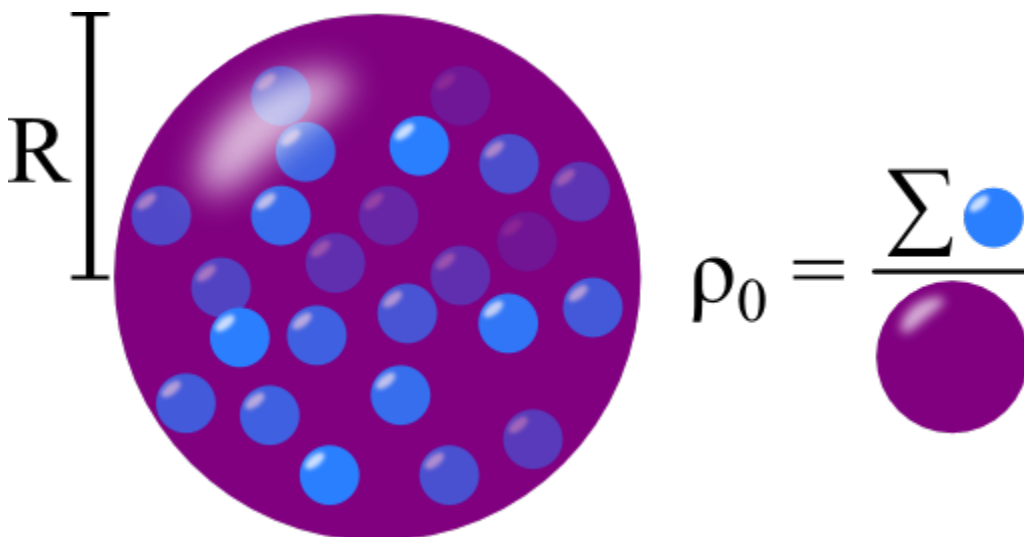
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To find the electrostatic potential of a general charge distribution can be quite complicated. However, if **symmetries** are present, the problem can be boiled down to the solution of much simple(r) equation(s). Let us employ the **spherical symmetry** of the homogeneously charged sphere to determine its potential and electric field!

Problem Statement



A homogeneously charged sphere of radius R is described by the **charge distribution**

$$\rho(\mathbf{r}) = \begin{cases} \rho_0 & r < R \\ 0 & r \geq R \end{cases}$$

The charge density ρ_0 can be understood as the number of homogeneously distributed charges vs. the volume of the sphere, see right figure.

Find the **electrostatic potential** of the homogeneously charged sphere and its **electric field**. You may want to follow these steps:

- Rewrite the **Laplace equation** for the electrostatic potential in a suitable coordinate system.
- **Simplify** the equation using the **spherical symmetry** of the charge distribution.
- Try to **integrate** the equation.
- Determine the **constants** of integration to find the **electrostatic potential**.
- Finally, calculate the **electric field**.

Hints

If a charge distribution is **spherically symmetric**, the same holds for the electrostatic potential, i.e. $\phi(\mathbf{r}) = \phi(r)$.

What is the **Laplace operator** in spherical coordinates?

Solution

Let us formally state the problem we have to solve in mathematical terms, i.e. relate the electrostatic potential and the given charge distribution via the **Laplace equation**:

$$\Delta\phi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\varepsilon_0}$$

$$\rho(\mathbf{r}) = \begin{cases} \rho_0 & r < R \\ 0 & r \geq R \end{cases}.$$

We notice that is an entirely radially symmetric charge distribution and hence the potential can only be a function of the radius. This symmetry will hugely simplify our calculations. Let us start with the field outside the sphere.

Electrostatic Potential outside the Sphere

Following **Gauss's law**, we know already that the field outside of the sphere has to be that of **a point charge**:

$$\oint_{\partial V} \mathbf{E}(\mathbf{r}) \cdot d\mathbf{A} = \frac{1}{\varepsilon_0} \int_V \rho(\mathbf{r}) dV$$

implies that for a charge distribution with spherical symmetry it does not matter how the charge inside the volume is distributed - only the magnitude of the charge counts. The magnitude is given by the **volume integral** over the whole charge density of the sphere. So we find the potential outside as

$$\begin{aligned} \phi(r) &= \frac{1}{4\pi\varepsilon_0} \frac{Q}{r} \text{ for } r \geq R \text{ with} \\ Q &= \int_0^{2\pi} \int_0^\pi \int_{r=0}^R \rho_0 dV' \\ &= \rho_0 \frac{4}{3} \pi R^3, \text{ so} \\ \phi(r) &= \frac{1}{3\varepsilon_0} \frac{\rho_0 R^3}{r} \text{ for } r \geq R. \end{aligned}$$

Ok, that was straight forward. We found the electrostatic potential outside of the sphere. Note how only one parameter - the **total charge Q of the sphere** - was important!

Solution inside of the Sphere

Now to the inner part which is a little more challenging. The **Laplace operator** in spherical coordinates is given by

$$\begin{aligned} \Delta f(r, \theta, \varphi) &= \frac{1}{r^2} \partial_r (r^2 \partial_r f(r, \theta, \varphi)) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta f(r, \theta, \varphi)) \\ &\quad + \frac{1}{r^2 \sin^2 \theta} \partial_{\varphi\varphi} f(r, \theta, \varphi). \end{aligned}$$

If now the potential is just a function of the radius, $\phi(\mathbf{r}) = \phi(r)$, we find that the Laplace equation inside of the sphere reduces to an **ordinary differential equation**:

$$\Delta \phi(r) = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \phi(r) \right) = -\frac{\rho_0}{\varepsilon_0}, \quad r < R.$$

We may solve this equation by **integration**. The first step yields

$$\int \frac{d}{dr} \left(r^2 \frac{d}{dr} \phi(r) \right) dr = - \int r^2 \frac{\rho_0}{\varepsilon_0} dr ,$$

$$r^2 \frac{d}{dr} \phi(r) = - \frac{1}{3} r^3 \frac{\rho_0}{\varepsilon_0} + c_1$$

where now c_1 is a **constant** we have to determine later. We proceed with the next integration:

$$\int \frac{d}{dr} \phi(r) dr = \int -\frac{1}{3} r \frac{\rho_0}{\varepsilon_0} + \frac{c_1}{r^2} dr ,$$

$$\phi(r) = -\frac{1}{6} r^2 \frac{\rho_0}{\varepsilon_0} - \frac{c_1}{r} + c_2 .$$

Now, we have to **determine the constants**. First of all we demand the potential to be finite at the origin, such that

$$c_1 = 0 .$$

Second, we also demand the potential to be continuous at the termination of the sphere:

$$\frac{1}{3\varepsilon_0} \frac{\rho_0 R^3}{R} \stackrel{!}{=} -\frac{1}{6} R^2 \frac{\rho_0}{\varepsilon_0} + c_2 ,$$

$$c_2 = \left(\frac{1}{3} + \frac{1}{6} = \frac{1}{2} \right) \frac{\rho_0 R^2}{\varepsilon_0} .$$

In the end we put everything together and find that for the homogeneously charged sphere the electrostatic potential is given by

$$\phi(r) = \begin{cases} \frac{1}{3\varepsilon_0} \frac{\rho_0 R^3}{r} & r \geq R \\ -\frac{1}{6} \frac{\rho_0}{\varepsilon_0} (r^2 - 3R^2) & r < R \end{cases} .$$

The Electric Field

The electrostatic potential varies only with respect to the radial coordinate. So, $\mathbf{E}(\mathbf{r}) = -\nabla\phi(r)$ can only have a component in radial direction. We can find the radial component by **differentiation** of the potential in the different areas:

$$E_r(r) = -\frac{d}{dr}\phi(r) = \begin{cases} \frac{1}{3\varepsilon_0} \frac{\rho_0 R^3}{r^2} & r \geq R \\ \frac{1}{3} \frac{\rho_0}{\varepsilon_0} r & r < R \end{cases} = \begin{cases} \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2} & r \geq R \\ \frac{1}{4\pi\varepsilon_0} \frac{Q}{R^3} r & r < R \end{cases}$$

To solve this problem we have made enormous use of the **rotational symmetry** of the problem.

We could boil down the **Laplace equation** to just an ordinary differential equation that could be solved by two successive integration steps.

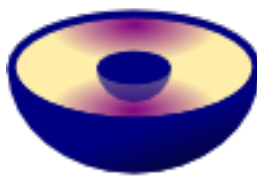
We ended up having some constants which could be found by

- checking if they are physically meaningful (finiteness) and
- using the boundary conditions.

We finally found the electric field by just one differentiation.

Interestingly the field outside of the homogeneously charged sphere was the same as a point charge. You probably have guessed that in the first place!

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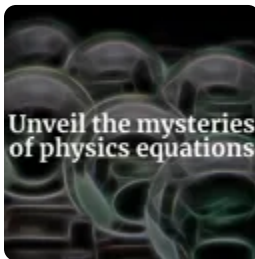
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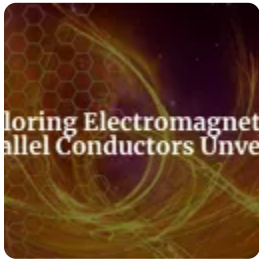
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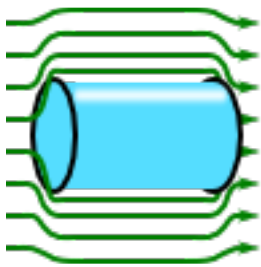
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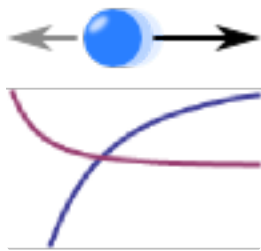
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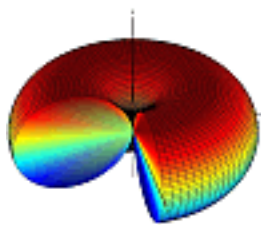
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